THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY
Department of Computer Science and Engineering

EventLens

Event Risk Research for Indices and Equities

Final Year Project Proposal

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Abstract

EventLens is a proposed research dashboard that connects prediction-market events to the risks faced by investors in major US indices and a small selection of popular equities. A user will be able to find relevant events, compare the market's probability with a wallet-informed estimate, explore possible portfolio outcomes, and ask an AI assistant to explain the supporting evidence.

The central research question is whether a wallet's earlier forecasting record adds useful information beyond Polymarket prices and overall trading activity. We will combine market data, public blockchain records and equity-price histories, using only information available at each historical forecast time. Simple probability models will be evaluated first, followed by event-based portfolio scenarios and volatility forecasts. The initial event family will be US Federal Reserve rate decisions, subject to a data-feasibility review.

The prototype will cover three index-tracking exchange-traded funds and five selected equities, with cash available for paper portfolio comparisons. AI agents will retrieve evidence, request calculations from tested services and explain results. Numerical estimates will remain traceable to their data and model versions. Historical testing, data quality and user comprehension will determine success; improved prediction or profitable trading will be tested rather than assumed.

Table of Contents

1 Introduction

1.1 Overview

Investors regularly face questions such as whether a central bank will change interest rates and how that decision might affect their holdings. News explains the event, price charts show market reactions, and prediction markets provide a changing estimate of the event's likelihood. Connecting these sources into a clear assessment of portfolio risk requires substantial data collection and specialist analysis.

Prediction-market prices are useful starting points, but their reliability can vary with liquidity and the participants trading in a market. Public wallet records offer another possible source of information: some participants may have a stronger forecasting record than others. A large position or a profitable trade alone, however, does not demonstrate forecasting skill. The value of wallet information must be measured against what the public price already reveals.

Even an accurate event probability does not directly tell an investor how much an equity might move. EventLens will make this second step explicit by linking selected events to historical asset responses or clearly labelled scenario assumptions. The interface will show both the estimated impact and the uncertainty in that mapping.

Research question

Does historically informative Polymarket wallet activity improve event-probability estimates beyond market prices and anonymous trading flow, and does any improvement help explain or forecast risk in selected indices and equities?

The project has three connected questions: whether wallet information improves probability forecasts, whether event information improves asset-risk forecasts, and whether an evidence-grounded AI interface helps users understand the results. Each question will be evaluated separately so that a useful interface does not conceal weak forecasting performance.

1.2 Objectives

1. **Build a trustworthy data foundation.** Join prediction-market definitions, prices, wallet activity and equity returns into a dataset that can reconstruct what was known at a particular time.
2. **Test a wallet-informed probability model.** Compare raw prices, calibrated prices, anonymous flow and wallet-aware flow using the same historical events.
3. **Explore event-related risk.** Estimate conditional portfolio outcomes and investigate whether event signals add value to conventional volatility forecasts.
4. **Deliver a readable research product.** Present sources, assumptions and results through a dashboard, scenario workspace and tool-using AI assistant.

The intended users are students, research-oriented retail investors and analysts who understand basic portfolio concepts but do not have their own blockchain and modelling infrastructure. The contribution is an evaluated connection from public event evidence to asset risk, supported by a reproducible product. It does not depend on discovering profitable traders or achieving positive investment returns.

Initial asset and event universe

The first release will use a fixed US equity universe, keeping calendars, currency treatment and data acquisition manageable. The following proposed list covers broad market exposure and familiar companies and will be frozen during the data audit.

Table 1: Proposed initial coverage. The ETF return is the portfolio input, not the index level.

Group	Coverage	Role
Key indices	S&P 500, Nasdaq-100 and Dow Jones Industrial Average	Research context, represented by SPY, QQQ and DIA respectively.
Popular equities	Apple (AAPL), Microsoft (MSFT), NVIDIA (NVDA), Amazon (AMZN), Tesla (TSLA)	A fixed list of widely followed equities.
Cash	USD cash balance	Reference allocation for paper comparisons; zero return over the short prototype horizon.
Events	Federal Reserve interest-rate decisions	One recurring family with explicit outcomes and documented announcement times.

This product-focused list is not representative of all equities. Retrospective results will acknowledge selection and survivorship limitations. Any replacement for insufficient data will be recorded before testing.

Minimum viable product

The MVP will support a watchlist and manually entered, long-only paper portfolios drawn from this universe. We will use one focal event per scenario, USD valuation and 1-, 5- and 20-trading-day horizons where data support them. Daily equity data are the minimum requirement; prediction-market snapshots will be collected prospectively at a configurable interval, initially targeting five minutes subject to access limits.

For each Federal Open Market Committee (FOMC) meeting, we will select one binary question, such as whether the target rate will be cut, and preserve its resolution rule. Contracts for the same meeting form one event group. Selection must match the user's horizon and the event definition, not merely the nearest expiry.

Extensions and exclusions

CPI releases and other macroeconomic events are conditional extensions if the primary sample is too small or the core pipeline is complete. Historical intraday volatility models, richer wallet models and a delta-hedged options paper backtest are also optional. They require separate evidence of adequate data and time.

Cryptocurrencies, foreign exchange, global coverage, unrestricted portfolios, identity inference, live orders and leverage are outside the MVP. Index options or VIX data will not substitute for equity-specific implied volatility.

1.3 Literature Survey

Prediction markets and informative participants

Prediction markets aggregate beliefs into tradable prices. Wolfers and Zitzewitz explain their forecasting value and the conditions under which a binary contract price can be interpreted as a probability [1, 2]. EventLens will therefore treat the market price as a strong baseline and test whether an adjustment improves its reliability. Calibration means that, across comparable cases, events assigned a given probability occur at approximately that frequency.

The Brier score measures the squared error of a probability forecast. Proper scoring rules, including Brier score and log loss, reward honest probabilities rather than confident guesses [3, 4]. They provide a more suitable test of an event forecast than trading profit, which is also affected by entry price, costs and market-making activity.

The 2026 working paper by Gómez-Cram and colleagues examines whether prediction-market accuracy is associated with an informed minority [5]. It motivates our wallet-level analysis but does not establish that the same effect will hold in our selected event family. Public addresses are not verified identities, and address clustering relies on assumptions [6]. We will study observed account wallets without claiming to identify people or insiders.

From event probability to asset risk

Event studies compare asset returns around information releases [7]. Their main difficulty here is separating the event's effect from other news and from expectations already reflected in prices. We will use historical response estimates cautiously and show sensitivity ranges when evidence is weak. An event probability is not, by itself, an equity-price forecast.

Portfolio risk has several meanings. Value at Risk (VaR) is a loss threshold at a chosen confidence level; expected shortfall, also called Conditional Value at Risk (CVaR), describes average loss in the worst tail [8]. Forecast volatility describes future variation, while realised volatility is measured from subsequent returns [9]. These quantities answer different user questions and will be labelled separately.

Option-implied volatility reflects option prices and need not equal the volatility expected under the actual return distribution. The implied–realised variance difference can contain a risk premium [10]. This is why an apparent variance gap is a research signal, not evidence of an arbitrage opportunity. Forecast comparisons will also account for noisy realised-volatility measurements [11].

AI assistance and the research gap

Retrieval-augmented generation and tool-using agents provide established patterns for connecting language models to external evidence [12, 13]. EventLens will use these patterns to retrieve events and explain model outputs, while deterministic software calculates probabilities and risk.

The project's specific contribution is to test the incremental value of persistent wallet information against anonymous flow, carry the comparison into a bounded equity-risk application, and expose the evidence through an understandable interface. It will connect established methods rather than claim that wallet skill, event studies or AI dashboards are themselves new discoveries.

2 Methodology

2.1 Design

A guided research journey

A user begins with an index, an equity or a small paper portfolio and selects a time horizon. EventLens then presents eligible events, explains why they may matter, and allows the user to inspect the probability evidence before opening a portfolio scenario. A contextual AI assistant can answer questions at each step. A saved report preserves the analysis for later review.

For example, a user researching SPY and MSFT over the next five trading days could inspect an upcoming rate decision, compare the market and adjusted probabilities, and explore a lower-exposure paper allocation. If the announcement lies outside the selected horizon or the equity-response estimate is unsupported, the interface will explain the limitation and withhold the affected calculation.

Table 2: Core interface features and the decisions they support.

Screen or feature	What the user sees	What the user can do
Overview and watchlist	Supported assets, upcoming eligible events, source freshness and coverage.	Select holdings, choose a horizon and open an event.
Event detail	Exact question, resolution rule, announcement time, raw and adjusted probability history.	Compare the estimates and inspect uncertainty and liquidity.
Wallet evidence	Historical sample size, directional flow, concentration and provisional skill status.	Examine why wallet activity changed the estimate without treating a ranking as certainty.
Scenario studio	YES/NO outcomes, portfolio loss estimates, VaR, expected shortfall and response assumptions.	Change a reviewed stress assumption or compare a bounded paper allocation.
Research assistant	An explanation tied to the selected event, portfolio, horizon and evidence.	Ask follow-up questions and open the supporting sources.
Saved research	Analysis date, inputs, model and scenario versions, source links and limitations.	Revisit a frozen run and export a readable report with a machine-readable result.

Design principles

The interface will use a black-and-white visual scheme with neutral grayscale accents, clear typography and consistent navigation. The event question and its implication for the selected assets take priority. Technical details, such as wallet-score components and model settings, appear in expandable evidence panels. Probability and loss charts will use labelled axes, legends and line styles so that interpretation does not rely on colour alone.

Every result will display its horizon, unit and as-of time. Observed prices, model estimates and user-entered assumptions will have distinct labels. Loading, missing data, stale data and model-unavailable states will be designed explicitly. A failed calculation will never appear as a zero-risk result. Keyboard navigation, readable contrast and accessible table alternatives will be included in the implementation.

The assistant remains next to the evidence being discussed. Selecting a cited result will open its source or calculation details; changing the horizon or portfolio will require a new calculation and visibly update the analysis context.

Concept screens

Figures 1 and 2 illustrate the intended layout. They are conceptual interface images; values are illustrative and do not represent research results.

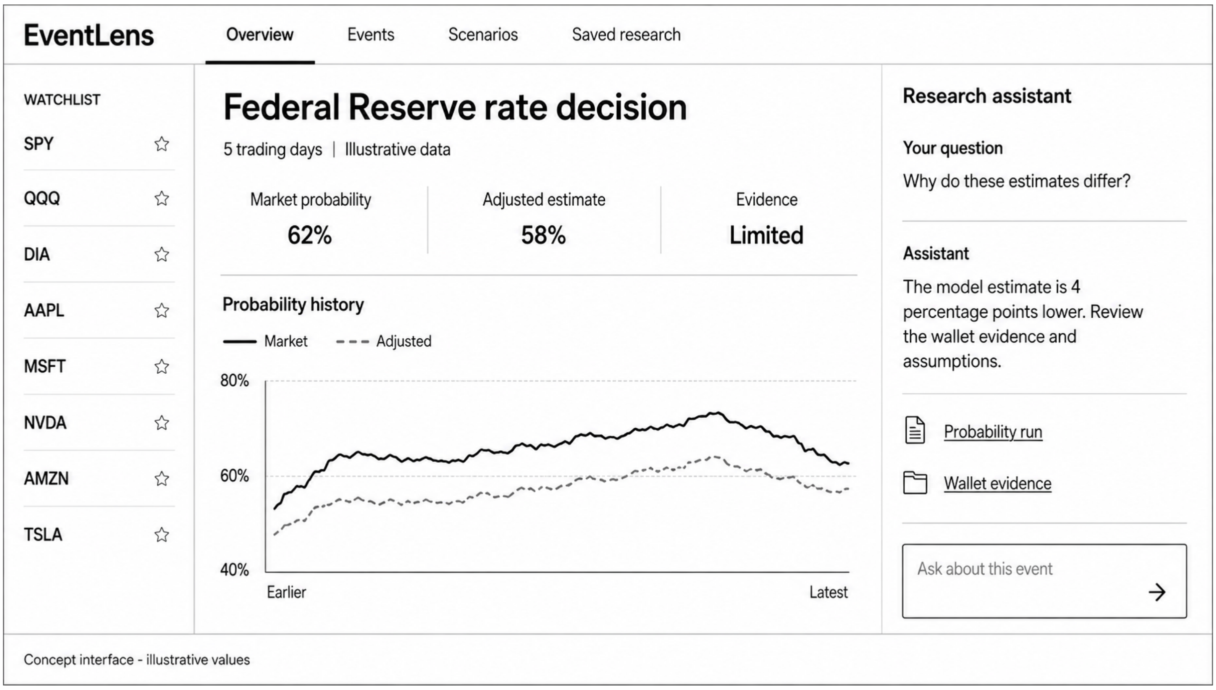

Figure 1: Overview concept: a watchlist, event probability comparison and contextual research assistant. All displayed values are illustrative.

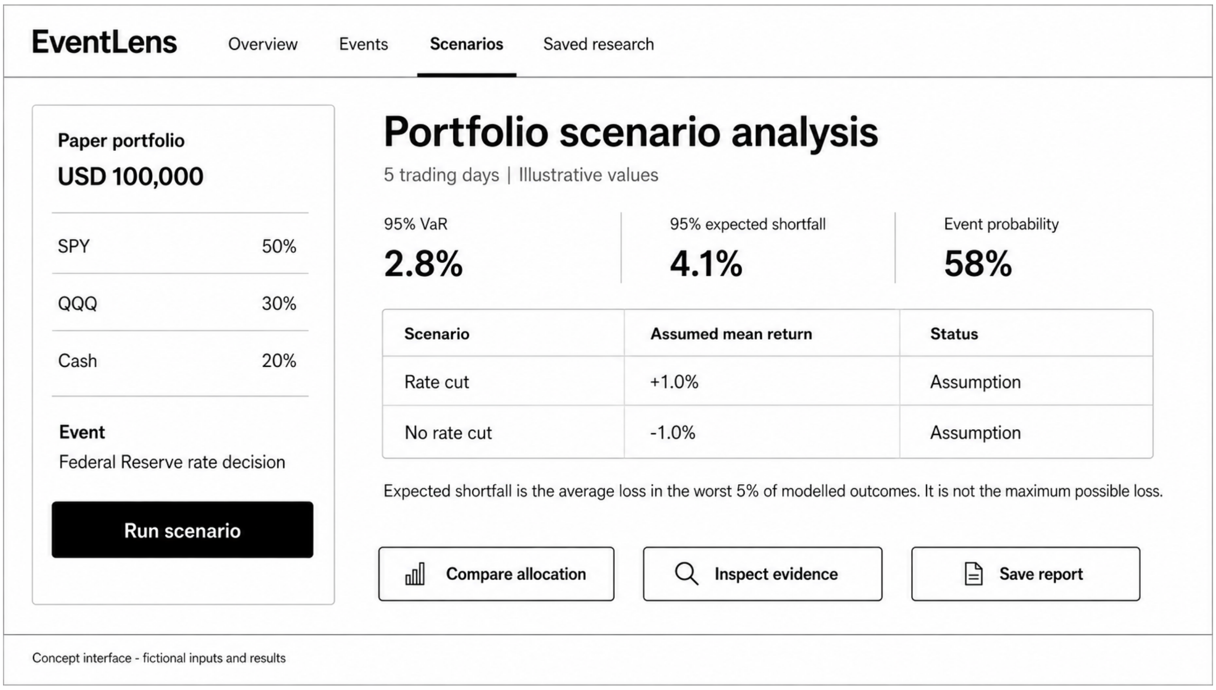

Figure 2: Scenario concept: portfolio inputs, clearly labelled model outputs and visible assumptions. The user compares paper scenarios without placing orders.

Three roles within one controlled workflow

The proposed assistant will use three logical roles: an **event research agent** to find relevant evidence, an **analysis coordinator** to request approved calculations, and an **explanation agent** to present results in context. These roles can share one model and orchestration service in the MVP. A separate autonomous agent infrastructure is unnecessary.

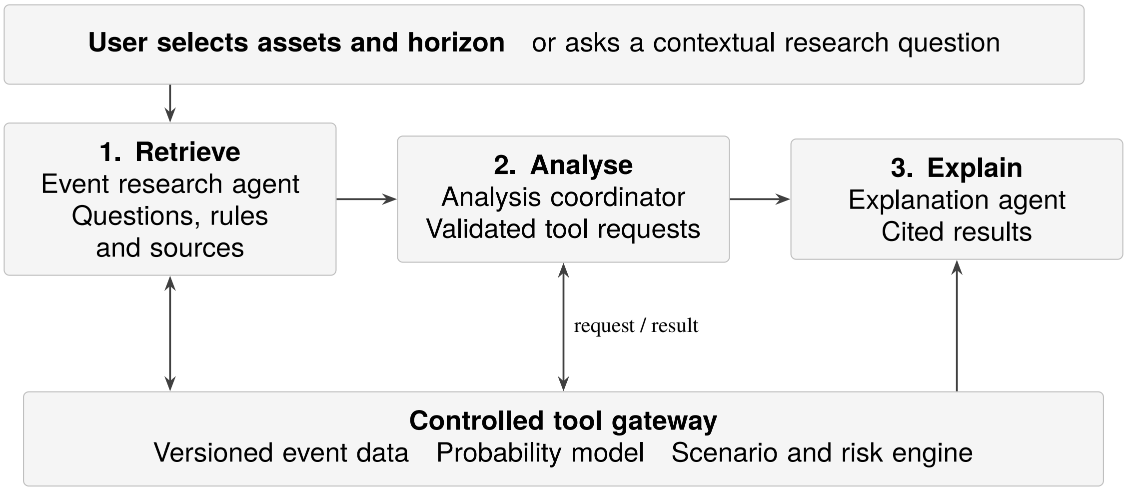

Figure 3: Proposed agent workflow. All quantitative results come from the same validated services used by the dashboard.

From a question to a supported answer

For "How could the next rate decision affect my SPY and MSFT exposure?", the assistant resolves the assets, horizon and analysis time. It retrieves an eligible event and proposes an event-to-asset explanation. Numerical responses must come from historical estimates or reviewed stress assumptions; generated text alone is not sufficient evidence.

The coordinator can call tools such as `find_events`, `get_wallet_evidence`, `get_probability`, `run_scenario` and `compare_allocations`. The gateway checks asset coverage, weights, units, data freshness and horizon compatibility. It returns structured values with sources, model versions and a research-run identifier. The explanation then links its numerical claims to those results and separates observations, estimates and assumptions [12, 13].

Boundaries and review

Users confirm changes to their paper inputs and can inspect assumptions before running a comparison. Researchers review new event-to-asset mappings and choose model versions using validation results. The agent cannot alter wallet labels, tune against the final test period, invent missing prices, calculate risk in prose or submit live orders. Market descriptions are treated as untrusted data, including any embedded instructions.

When evidence is incomplete, the assistant will identify the gap and offer a supported view, such as the raw probability. Tool traces and saved reports make answers reviewable. Evaluation will cover citations, numerical agreement, abstention, user understanding, response time and cost.

Sources and their roles

The data pipeline will combine three kinds of evidence: prediction-market information, public blockchain settlement records and conventional financial data. These sources serve different purposes. Polymarket matches orders off-chain and settles trades on-chain, so an order-book snapshot is not itself a blockchain record [15].

Table 3: Planned sources, with access decisions made during the feasibility audit.

Source	Data to acquire	Use and limitation
Polymarket metadata API (Gamma)	Event definitions, rules, identifiers, outcome prices and available price history.	Define the forecast target; archive changes to mutable metadata [16].
Polymarket CLOB and activity Data API	Bid/ask snapshots, liquidity and public wallet activity.	Measure market quality and candidate wallet flow; historical book depth may need prospective collection.
Goldsky Polymarket datasets	Indexed fills, matched trades and relevant position records.	Targeted historical reconstruction, subject to version coverage and access cost [17].
Polygon RPC	Transaction receipts, contract logs and block references.	Verify a stratified sample of settlements; older state may require archival access.
Alpha Vantage or an approved equivalent	Daily equity/ETF prices, splits and dividends; intraday data only if available.	Proposed primary price-provider adapter. Adjusted daily data are a premium function; coverage and academic-use terms must be checked [18].
Federal Reserve	Meeting dates, statements and announced rate decisions.	Establish announcement timing and validate event labels separately from platform resolution [19].

The initial audit will test real sample downloads for all eight securities and the chosen event family before fixing the provider. If access is insufficient, the team will use a documented, permitted university dataset or equivalent licensed export through the same schema. Provider, adjustment convention and coverage will be fixed for each experiment; overlapping series will be compared before any switch.

A bounded collection strategy

We will enumerate eligible events, collect their public metadata, and backfill only relevant market and wallet histories. We will not mirror the entire blockchain or all Polymarket positions. Goldsky documents a 2026 contract migration and warns that legacy public subgraphs may be incomplete; the audit must therefore verify version-specific coverage rather than assume one continuous dataset [17].

Forward market snapshots will be retained from the start of the project, while equity bars will be refreshed after the trading session. Historical options quotes are a separate acquisition decision and are not required for the core dashboard. No provider subscription or historical depth is assumed to have been secured by this proposal.

Turning source records into research inputs

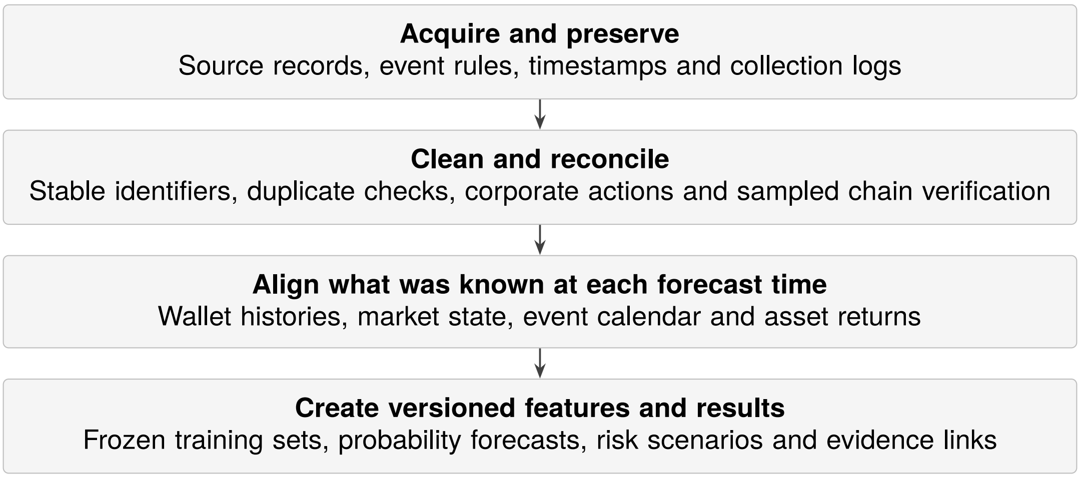

Figure 4: Processing pipeline. Raw evidence is retained so each transformation can be checked and repeated.

Preserve identifiers and time. Each event is linked to its markets, outcome tokens and settlement records. Raw records retain source time, collection time and source version. Announcement time, platform resolution time and data-availability time remain separate. Historical records downloaded later are admissible only when their content can be shown to have existed at the forecast cut-off; mutable current rankings are not historical features.

Reconstruct activity consistently. Wallet analysis retains both participant perspectives, while market volume counts the economic trade once. A transaction may contain several logs, so transaction hash alone is not a unique record key. We will use chain, transaction, log and contract identifiers and verify sampled amounts against Polygon receipts. Position reconstruction must account for transfers, splits, merges and redemptions as well as trades. Incomplete positions receive a quality flag.

Align asset returns. Timestamps will be normalised to UTC while retaining the US trading calendar, daylight-saving changes and after-hours announcements. Splits and dividends will be handled consistently. Future adjustments or revised data must not change historical features without a recorded version. Missing bars will be flagged rather than filled with zero returns; stale probabilities will not be carried across an announcement as if newly observed.

Make processing reproducible. Immutable source snapshots and checksums will accompany cleaned tables and Parquet research files. Collectors will use retries, rate limits and checkpoints. Replaying the same batch must not duplicate records. Each model run stores the data manifest, configuration, code version and random seed, and the interface will expose coverage gaps through its evidence panel.

Estimate the event probability

We will begin with a raw probability based on the YES bid/ask midpoint, recording spread, depth and timestamp. When both outcome books are usable, their midpoints can be normalised for consistency. A missing or one-sided book will be flagged, not silently replaced by an unrelated latest trade.

A simple regularised logistic model will calibrate this estimate using earlier resolved events. The wallet-informed model will then add a small number of signals: overall directional flow, the flow of historically informative wallets and market-quality controls. Coefficients will be fitted on training data, rather than chosen to produce a desired adjustment. Beta calibration is a possible robustness check if justified by the sample [14].

Assess wallet information conservatively

An informative wallet is an observed account whose earlier decisions contain information beyond the contemporaneous market price. Our first score will examine whether the wallet repeatedly chose the eventual outcome at informative prices, whether prices subsequently moved in its direction, and how broad its prior history was. Returns after fees will be a supporting diagnostic; profitable liquidity provision alone is not proof of forecasting skill.

Repeated fills will be consolidated into wallet–event decisions. Balanced YES and NO holdings will be distinguished from directional exposure. Scores based on few independent events will be pulled towards the population average, a technique called shrinkage. A provisional eligibility rule is ten earlier resolved events across at least three announcement dates, with five- and twenty-event thresholds checked in sensitivity analysis. If the history cannot support this rule, the wallet signal will be unavailable rather than assigned artificial confidence.

Current contributions will combine earlier skill, direction, recency and capped position size. New-wallet anomalies remain separate from historical skill. No single large wallet should dominate the adjustment. A learned hierarchical wallet model is an extension, conditional on enough independent events; the initial transparent rule model remains useful for testing feasibility.

Link events to assets

The event research agent may suggest that a rate decision matters to SPY or MSFT, but numerical impacts will come from matched historical event windows and conventional market controls. The mapping must account for surprise relative to the expectation already embedded in prices. Sparse asset-specific estimates will be pooled or replaced by explicitly reviewed stress ranges.

Wallet information may improve an event probability without improving an equity-risk forecast. The event-to-asset response is a separate estimate with its own uncertainty and evidence requirements.

Each mapping will retain its event definition, horizon, sample size, estimated response range and review status. A terminal YES/NO mixture will only be used when the economic announcement falls within the risk horizon. Otherwise the MVP will offer a compatible horizon or withhold that calculation. Correlated contracts from one meeting will not be treated as independent shocks.

Explore portfolio scenarios and tail risk

For a selected portfolio, the engine will combine the conditional return distributions for an event occurring (YES) and not occurring (NO). Let p be the selected event probability, μ a mean portfolio return and σ^2 its variance over the same horizon. The key calculations are:

$$\mu_{\text{mix}} = p\mu_{\text{YES}} + (1-p)\mu_{\text{NO}}, \tag{1}$$
$$\sigma^2_{\text{mix}} = p\sigma^2_{\text{YES}} + (1-p)\sigma^2_{\text{NO}} + p(1-p)(\mu_{\text{YES}} - \mu_{\text{NO}})^2. \tag{2}$$

The last term captures uncertainty about which outcome occurs. Consequently, a higher adverse-event probability can worsen expected loss without necessarily increasing variance. For multiple holdings, we will preserve cross-asset dependence when sampling returns, then apply the user's weights to each joint scenario.

The engine will report scenario returns, horizon volatility, a chosen loss-threshold probability, 95% VaR and 95% expected shortfall. VaR is a threshold, not the maximum loss. Expected shortfall will be calculated as the average of the worst 5% of the weighted loss distribution, including fractional threshold mass where needed. We will use empirical or heavy-tailed scenarios and check sensitivity to the probability and asset-response assumptions.

Paper comparisons will hold the event and model version constant while comparing the current allocation with a user-selected reduction in a supported holding and a corresponding increase in cash. Weights must sum to one and remain non-negative. Any displayed benefit must be considered alongside turnover, transaction-cost assumptions and estimation uncertainty.

Test whether event signals improve volatility forecasts

The core forecasting experiment will compare a conventional daily-return volatility baseline with the same baseline augmented by event probabilities, their recent changes and the difference between adjusted and raw estimates. Rolling historical variance and exponentially weighted variance provide simple benchmarks. The initial observed target over H trading days will be the sum of squared daily log returns:

$$RV_{t,H} = \sum_{h=1}^{H} r^2_{t+h}. \tag{3}$$

This is a noisy daily-data proxy for realised variation; at one day it is especially noisy. It differs from the variance of the terminal portfolio return above. We will compare like-for-like targets and annualisation conventions and will not insert the terminal mixture's between-outcome term into a path-variance formula. If suitable intraday histories exist, a richer realised-variance measure and HAR-RV, GARCH-X or a tree model can be explored. These are candidate methods, not commitments to implement all models.

Optional volatility trading study

A delta-hedged paper options backtest may compare forecast variance with option-implied variance only after acquiring historical quotes with matching underlying, horizon and expiry. Costs, bid/ask spreads, hedge frequency and execution assumptions must be included. Results would report after-cost profit and loss, drawdown and sensitivity rather than promise a variance premium. This extension is optional and will not displace the core probability, risk or interface evaluation.

2.2 Implementation

We propose a modular Python backend and a TypeScript web frontend. One deployable application with a background worker will keep the four-person project manageable while separating collection, modelling, risk calculation and AI explanation.

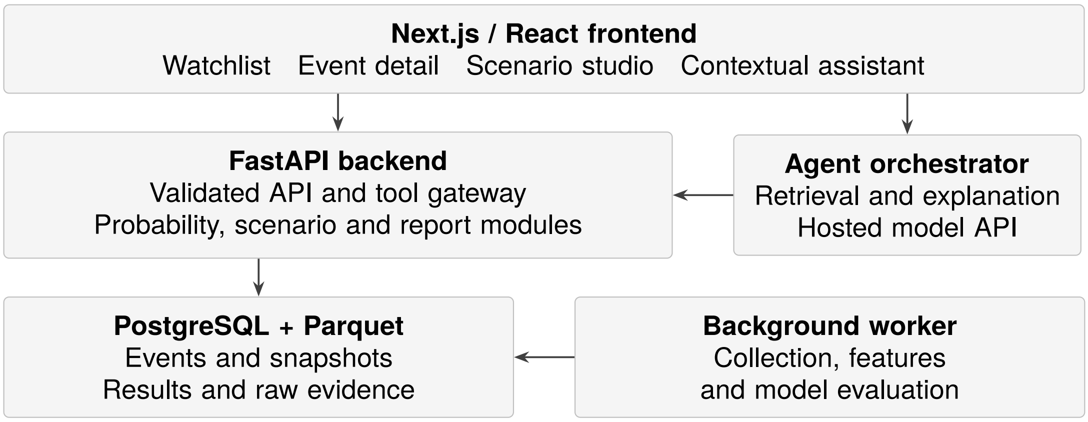

Figure 5: Proposed application architecture. The dashboard and assistant share the same calculation services and stored results.

Table 4: Preferred choices and the reasons for them.

Layer	Choice	Reason and trade-off
Frontend	Next.js, React, TypeScript	Reusable charts and forms; client components for interaction and server components for page/data composition [21].
Backend	FastAPI and Pydantic	Keep numerical work in Python; typed validation and OpenAPI contracts support both UI and agent tools [20].
Research	Polars/Pandas, NumPy, scikit-learn; statsmodels as needed	Familiar data and statistical tooling; begin with interpretable models.
Storage	PostgreSQL and Parquet	Relational identifiers and run metadata alongside efficient research files. TimescaleDB is optional if measured volume justifies it.
Jobs and deployment	Python worker, Docker Compose	Move long ingestion/training jobs out of request handling; use one shared deployment before considering distributed services.

The frontend will receive model outputs, units, quality flags and run identifiers from the backend rather than reimplement financial calculations. Long analyses return a job identifier with queued, running, completed or failed status. Cached results are keyed by data, model, scenario, portfolio and horizon so a changed input cannot silently reuse an incompatible result.

The deployment will use authenticated team access, server-side secrets, persistent database and raw-data volumes, and scheduled backups. An API-hosted language model avoids local model training; a small evaluation will select the provider based on grounded-answer quality, latency and cost. Model versions and prompts will be recorded. The dashboard will remain usable for direct research when the assistant is unavailable.

2.3 Testing

The testing programme will cover the data pipeline, numerical services and complete user workflow. Ingestion tests will check schema validation, duplicate-free replay, interrupted collection and reconciliation against sampled blockchain records. Point-in-time checks will reject features or wallet histories that include information published after the forecast cut-off.

Risk calculations will be checked against reference cases, including event probabilities of zero and one, missing assets, extreme losses and singular covariance estimates. API tests will enforce supported assets, compatible horizons and non-negative portfolio weights that sum to one. The frontend and assistant must display the same versioned results.

End-to-end tests will cover event selection, scenario comparison and report retrieval, including unavailable or stale data. Agent tests will include malformed tool requests and hostile instructions in retrieved text. Every numerical claim must trace to a saved result; no test may produce a live order or violate allocation constraints.

2.4 Evaluation

Experimental design

The audit will target 20–50 resolved markets, if recoverable, and report independent announcement dates separately. Several contracts or snapshots of one FOMC meeting do not create additional outcomes. If the sample cannot support reliable fitting, we will report a descriptive pilot. A broader macro sample requires a documented scope revision before testing.

Training, validation and testing will follow chronological order, with each event's contracts and snapshots kept together. Overlapping return windows will be purged at split boundaries. Wallet scores, normalisation and scenarios must use only pre-cut-off information. The final test interval stays unopened until models and thresholds are fixed.

Table 5: Baseline comparisons isolate the contribution of each information layer.

Model	Information used	Question tested
B0	Historical event frequency or conventional asset variables	What is possible without prediction-market signals?
B1	Raw market probability	What does the public price already tell us?
B2	Calibrated price and liquidity	Does basic market-quality adjustment help?
B3	B2 plus anonymous directional flow	Does overall trading activity add information?
B4	B3 plus past-only wallet-informed flow	Does persistent wallet information add value?
Optional	Learned wallet weights or richer models	Is extra complexity justified by held-out results?

Performance measures

Probability evaluation will use Brier score, log loss and reliability plots. Direction randomisation, identity shuffles, concentration controls and eligibility-threshold sensitivity will test apparent wallet skill. We will report event-level confidence intervals and small-sample limitations. Multiple wallet-level tests require false-discovery control.

Volatility forecasts will use variance error and QLIKE, a variance-forecast loss function [11]. Portfolio diagnostics include VaR exceedances and exploratory expected-shortfall checks. Probability variants will share the same asset-response model to isolate the adjustment's downstream effect. Strong claims require enough independent observations.

We will target 6–8 participants for a formative study: find an event, explain a probability difference, compare scenarios and identify a limitation. We will record completion, time, errors and feedback. Fixed questions will compare the agent with the dashboard alone and a retrieval-only assistant, measuring source coverage, numerical agreement, abstention, latency and cost.

Success criteria and limitations

Success requires reproducible data, controlled comparisons, a tested risk engine and a usable evidence-linked interface. Positive, negative and null findings will be reported; profitability is not required.

Table 6: Principal risks and planned responses.

Risk	Response
Sparse independent events	Report event counts; use a descriptive pilot or simple model; broaden the event family only through a documented revision.
Missing or costly history	Audit access early, backfill selectively and collect prospectively; disable unsupported features.
Selection bias and hidden hedges	Freeze the universe, reconstruct participants historically, shrink wallet scores and avoid claims about real-world identity or total conviction.
Confounded asset responses	Use controls, reviewed scenario ranges and sensitivity analysis; distinguish stress assumptions from validated forecasts.
Protocol or API changes	Version source schemas and contracts, test adapters and reconcile sampled settlements.
Unsupported AI answers	Restrict tools, validate arguments, link claims to results and test missing-data abstention and prompt injection.
Time and integration pressure	Integrate an early vertical slice; defer complex models and options trading until the core evaluation is complete.

The final package will contain a documented dataset and acquisition scripts, model and baseline experiments, a tested scenario/risk service, the working dashboard and assistant, and a report covering research findings and usability. Data redistribution will follow provider permissions; where raw data cannot be shared, the package will provide acquisition instructions, schemas and permitted sample fixtures. The demonstration will include both a supported analysis and a case with insufficient data.

3 Project Planning

3.1 Division of Work

Franklin, Anthony, Owen and Sunny will work across four tracks, each with a lead and a reviewer. Named ownership remains to be agreed. All members will contribute to literature review, reporting and presentations.

Table 7: Proposed work allocation, with named ownership to be recorded by the team.

Track	Responsibility	Deliverable
Data and blockchain	Source adapters, collection, reconciliation and quality reporting.	Versioned dataset and coverage report.
Probability research	Wallet scoring, calibration, baseline comparisons and leakage controls.	Reproducible probability experiments.
Scenarios and risk	Asset-response estimates, volatility, tail risk and reference tests.	Deterministic risk engine and application study.
Product and AI	Backend integration, frontend, agent tools and deployment.	Integrated dashboard and user evaluation.

Git will hold code, experiment configuration and separate LaTeX section files for collaborative editing. Generated files and secrets stay out of version control. Weekly reviews and an issue tracker will record owners, dependencies, meeting decisions and experiment changes.

3.2 GANTT Chart

The schedule is relative to project commencement and will be aligned with course dates. Data feasibility comes first; interface work starts with sample data to identify integration problems early.

Table 8: Planning baseline. Shaded cells indicate the main work period.

Workstream	1–4	5–8	9–12	13–16	17–20	21–24
Scope, literature and feasibility						
Collection and reconciliation						
Wallet and probability models						
Event mapping and risk engine						
Interface and AI integration						
Evaluation and user study						
Report and reproducibility						

Milestones. Week 2: initial access and scope decision. Week 7: a versioned dataset and reconciliation report. Week 12: first raw-price, anonymous-flow and wallet-flow comparison. Week 18: working scenarios and risk calculations. Week 22: integrated evaluation and user tasks. Week 24: final report, demonstration and reproducibility package. Failed feasibility gates will narrow the model or coverage before optional work begins.

4 Hardware and Software Requirements

4.1 Hardware Requirements

The initial models and API-hosted language model do not require a local GPU. Development can be carried out on standard personal computers, with a modest shared server for scheduled collection and the demonstration.

Table 9: Proposed hardware provision.

Resource	Minimum provision	Preferred provision
Developer computer	Four-core CPU and 8 GB RAM	16 GB RAM for concurrent local services.
Storage	30 GB free disk space	Approximately 100 GB SSD space for bounded historical data.
Shared server	Local demonstration if necessary	2–4 vCPU and 8–16 GB RAM, subject to measured workload.
Network	Stable internet connection	Reliable access for collectors, hosted models and remote collaboration.

A self-hosted Polygon node is outside the initial scope. Hosted RPC and targeted indexed data will be used for collection and verification. Database and raw-data storage must persist across application updates and be backed up regularly.

4.2 Software Requirements

Table 10: Required software and external services.

Category	Software or service	Purpose
Runtime	Python and Node.js	Research services, collectors and web application.
Application	FastAPI, Pydantic, Next.js, React and TypeScript	Validated backend APIs and interactive interface.
Data and modelling	PostgreSQL, Parquet, Polars/Pandas, NumPy and scikit-learn	Storage, processing and initial statistical models.
External services	Polymarket APIs, Goldsky, Polygon RPC, equity data and hosted LLM	Event evidence, asset histories and research assistance.
Quality and deployment	Pytest, browser tests, Docker and Docker Compose	Automated verification and reproducible deployment.
Collaboration	Git, issue tracker and LaTeX	Source control, task allocation and shared documentation.

External costs may include adjusted equity histories, indexed blockchain access, archival RPC, a hosted language model and server hosting. The team will set a spending cap after checking sample data and provider terms during the feasibility stage. A paid historical options dataset is not required for the core project. Software dependencies, model versions and configuration will be recorded to support reproducibility.

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6 Appendix A: Meeting Minutes

Initial project meeting

Date	8 August 2026
Location	Hong Kong
Attendees	Franklin, Anthony, Owen and Sunny
Purpose	Discuss the project direction, scope, evaluation approach and division of work.
Status	Preliminary discussion; formal scope decisions and individual ownership remained open.

Discussion summary

The meeting considered a Web3 quantitative-trading and market-analysis agent. Subsequent research narrowed the proposed direction to Polymarket, public wallet activity, event-conditioned volatility and portfolio risk. The present proposal focuses the initial application on selected indices and popular US equities, with options paper trading retained as a possible extension.

Matters requiring confirmation

The team will confirm the initial asset and event universe, named workstream leads, data access and the schedule against official course milestones. Decisions and action items will be recorded with owners and deadlines at the next meeting.

Formal scope approval and named ownership were not recorded at this meeting.